CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Ratio and Proportion

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2+3x-5}{3x-5}=\frac{9x^2+2x+5}{2x+5}$, where $x \neq 0$. ICSE 2025, 3 marks

SOLUTION

Apply dividendo: if $\frac{a}{b}=\frac{c}{d}$ then $\frac{a-b}{b}=\frac{c-d}{d}$.

$\frac{6x^2+3x-5-(3x-5)}{3x-5}=\frac{9x^2+2x+5-(2x+5)}{2x+5}$, so $\frac{6x^2}{3x-5}=\frac{9x^2}{2x+5}$

Since $x \neq 0$, divide both sides by $3x^2$: $\frac{2}{3x-5}=\frac{3}{2x+5}$

$4x+10=9x-15$, so $5x=25$.

Answer: $x=5$

2. Given $\frac{(a+b)^3}{(a-b)^3}=\frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a:b$, using the properties of proportion.

ICSE 2024, 4 marks

SOLUTION

(a) Take the cube root of both sides: $\frac{a+b}{a-b}=\frac{4}{3}$

(b) Apply componendo and dividendo: $\frac{(a+b)+(a-b)}{(a+b)-(a-b)}=\frac{4+3}{4-3}$

$\frac{2a}{2b}=\frac{7}{1}$

Answer: (a) $\frac{4}{3}$ (b) $a:b=7:1$

3. Use componendo and dividendo to solve $\frac{\sqrt{2x+2}+\sqrt{2x-1}}{\sqrt{2x+2}-\sqrt{2x-1}}=3$. ICSE 2023, 3 marks

SOLUTION

Write 3 as $\frac{3}{1}$ and apply componendo and dividendo: $\frac{2\sqrt{2x+2}}{2\sqrt{2x-1}}=\frac{3+1}{3-1}=2$

Square both sides: $\frac{2x+2}{2x-1}=4$, so $2x+2=8x-4$

$6x=6$

Answer: $x=1$

4. Find $x : y$, given $\frac{x^2+2x}{2x+4} = \frac{y^2+3y}{3y+9}$. ICSE 2020, 3 marks

SOLUTION

Factorise: $\frac{x(x+2)}{2(x+2)} = \frac{y(y+3)}{3(y+3)}$

$\frac{x}{2} = \frac{y}{3}$. By alternendo, $\frac{x}{y} = \frac{2}{3}$.

Answer: $x : y = 2 : 3$

5. If $x = \frac{\sqrt{2a+1}+\sqrt{2a-1}}{\sqrt{2a+1}-\sqrt{2a-1}}$, prove that $x^2 - 4ax + 1 = 0$. ICSE 2020, 4 marks

SOLUTION

Write x as $\frac{x}{1}$ and apply componendo and dividendo: $\frac{x+1}{x-1} = \frac{2\sqrt{2a+1}}{2\sqrt{2a-1}}$

Square both sides: $\frac{x^2+2x+1}{x^2-2x+1} = \frac{2a+1}{2a-1}$

Apply componendo and dividendo again: $\frac{2x^2+2}{4x} = \frac{4a}{2}$, so $\frac{x^2+1}{2x} = 2a$

$x^2 + 1 = 4ax$, that is $x^2 - 4ax + 1 = 0$. **Proved.**

6. Use the properties of proportion to solve $\frac{\sqrt{5x}+\sqrt{2x-6}}{\sqrt{5x}-\sqrt{2x-6}} = 4$. ICSE 2019, 3 marks

SOLUTION

Apply componendo and dividendo: $\frac{2\sqrt{5x}}{2\sqrt{2x-6}} = \frac{4+1}{4-1} = \frac{5}{3}$

Square both sides: $\frac{5x}{2x-6} = \frac{25}{9}$, so $45x = 50x - 150$

Answer: $x = 30$

7. Use the properties of proportion to solve $\frac{2x+\sqrt{4x^2-1}}{2x-\sqrt{4x^2-1}} = 4$, given that x is positive. ICSE 2018, 3 marks

SOLUTION

Apply componendo and dividendo: $\frac{4x}{2\sqrt{4x^2-1}} = \frac{4+1}{4-1}$, so $\frac{2x}{\sqrt{4x^2-1}} = \frac{5}{3}$

Square both sides: $\frac{4x^2}{4x^2-1} = \frac{25}{9}$, so $36x^2 = 100x^2 - 25$

$64x^2 = 25$, and x is positive.

Answer: $x = \frac{5}{8}$

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8. Given $\frac{7m+2n}{7m-2n}=\frac{5}{3}$, use the properties of proportion to find (a) $m:n$ (b) $\frac{m^2+n^2}{m^2-n^2}$. ICSE 2017, 3 marks

SOLUTION

(a) Apply componendo and dividendo: $\frac{14m}{4n}=\frac{5+3}{5-3}=4$, so $\frac{m}{n}=\frac{8}{7}$.

(b) $\frac{m^2}{n^2}=\frac{64}{49}$. Apply componendo and dividendo: $\frac{m^2+n^2}{m^2-n^2}=\frac{64+49}{64-49}$

Answer: (a) $8:7$ **(b)** $\frac{113}{15}$

2 Proportion

9. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y. (a) $x=4, y=150$ (b) $x=3, y=100$ (c) $x=4, y=100$ (d) $x=3, y=150$ ICSE 2024, 1 mark

SOLUTION

Speed $=\frac{60}{2}=30$ m/s. At uniform speed, distance is proportional to time.

$x=\frac{90}{30}=3$ and $y=30\times 5=150$

Answer: (d)

10. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion? ICSE 2023, 3 marks

SOLUTION

Let the number be x. Then $\frac{4+x}{6+x}=\frac{8+x}{11+x}$.

$(4+x)(11+x)=(6+x)(8+x)$, so $44+15x+x^2=48+14x+x^2$

$x=4$. Check: 8, 10, 12, 15 gives $8\times 15=10\times 12=120$.

Answer: 4

11. The numbers $K+3$, $K+2$, $3K-7$ and $2K-3$ are in proportion. Find K. ICSE 2019, 3 marks

SOLUTION

$\frac{K+3}{K+2}=\frac{3K-7}{2K-3}$, so $(K+3)(2K-3)=(K+2)(3K-7)$

$2K^2+3K-9=3K^2-K-14$, so $K^2-4K-5=0$

$(K-5)(K+1)=0$

Answer: $K=5$ **or** $K=-1$

3 Continued proportion

12. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 ICSE 2023, 1 mark

SOLUTION

Mean proportional $= \sqrt{4 \times 9} = \sqrt{36} = 6$

Answer: (b)

13. b is the mean proportional between a and c. Show that $\frac{a^4 + a^2b^2 + b^4}{b^4 + b^2c^2 + c^4} = \frac{a^2}{c^2}$. ICSE 2017, 3 marks

SOLUTION

b is the mean proportional, so $b^2 = ac$.

Numerator $= a^4 + a^2(ac) + (ac)^2 = a^2(a^2 + ac + c^2)$

Denominator $= (ac)^2 + (ac)c^2 + c^4 = c^2(a^2 + ac + c^2)$

LHS $= \frac{a^2(a^2 + ac + c^2)}{c^2(a^2 + ac + c^2)} = \frac{a^2}{c^2}$ = RHS. **Proved.**

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